Flywheel rotor design: from the optimal shape to a manufactured part

Project report

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Abstract

This report covers the whole project. A first Arduino motor bench showed the effect of a rotating mass, without being conclusive. The TIPE (2024–2025), a research project carried out during the French *classes préparatoires*, looked for the shape that maximises inertia and modelled the braking of a vehicle, with a ramp experiment. The project then proves the shape result, adds the constraint that was missing (material strength), checks the resulting profiles in 2D, then adapts them to 3D printing and models them in CAD and finite elements. Each section separates what is measured from what is computed.

1 Introduction

The starting question is the TIPE's: can a flywheel store part of the kinetic energy a vehicle dissipates when braking, without degrading braking? This is the principle of flywheel kinetic energy recovery systems studied for cars [5]. The question splits in two:

1. which flywheel shape stores the most energy;
2. how to integrate it with braking without degrading performance.

Table 1 gives the thread of the project.

Step	Question	Contribution
Preliminary test (Arduino)	Effect of a rotating mass on a motor	Inertia visible, losses too high, measurement too coarse
TIPE (2024–2025)	Maximum-inertia shape; braking with and without a flywheel	Circle at fixed perimeter; deceleration model consistent with measurement
Part 1	Prove the shape result	Theorems (Wirtinger, annulus)
Part 2	Shape that stores the most energy *before failure*	Profiled disk, checked in 2D and with CalculiX
Part 3	Does the shape survive manufacturing?	Printable parts, sensitivity to printing defects
CAD and simulation	Move to industrial tools	Fusion 360 parts, Abaqus input files

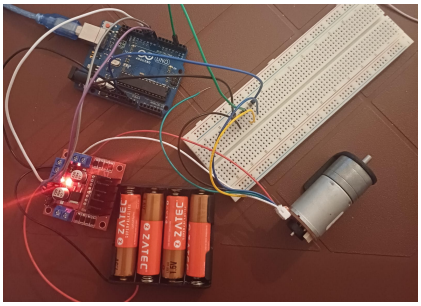
Table 1: Project steps.

2 Preliminary test: Arduino motor bench

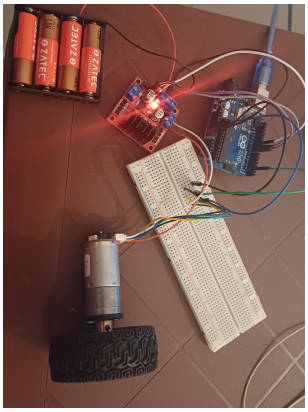
2.1 Set-up

The bench uses an Arduino Uno board, an L298N driver, a JGA25-370 geared motor with a Hall-effect encoder, and a 12 V battery pack. A voltage step is applied for 10 s, then cut, and the speed

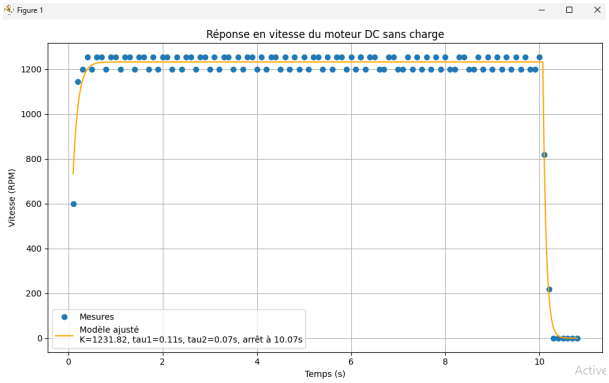
is read from the encoder. Two cases are studied: motor alone, and motor driving a wheel (figure 1). A first-order model is fitted to the spin-up (time constant τ_1) and to the spin-down (time constant τ_2).



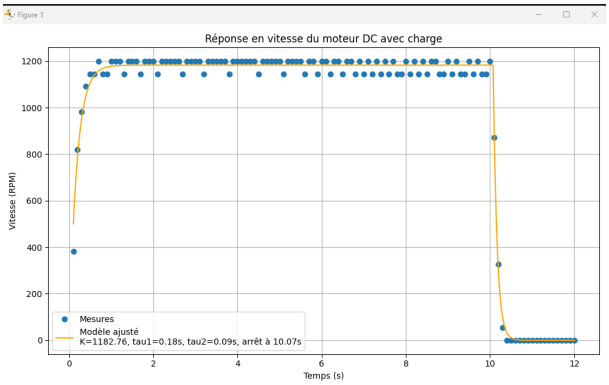
(a) Set-up without load



(b) Set-up with wheel



(c) Response without load



(d) Response with wheel

Figure 1: Motor bench: set-up and speed responses (measured points, fitted model; original plot labels in French).

2.2 Results and discussion

	No load	With wheel	Change
Steady speed (rpm)	1232	1183	−4 %
Spin-up time constant τ_1 (s)	0.11	0.18	+64 %
Spin-down time constant τ_2 (s)	0.07	0.09	+29 %

Table 2: Fitted parameters (read from the plots).

The wheel clearly lengthens the spin-up: this is the expected effect of inertia. If τ_1 is taken as proportional to the total inertia (friction neglected), the wheel would add about 64 % to the motor's reflected inertia. This estimate is debatable.

The result is limited in three ways:

- **Energy is not retained**: the system stops in 0.2 to 0.3 s ($3\tau_2$). The gearbox and the driver dissipate almost everything, so a geared motor with a wheel is a poor flywheel.
- **The measurement is coarse**: the points jump between two levels (about 1145 and 1200 rpm on the with-wheel plot, about 1200 and 1255 without load), a resolution of about 5 %, due to

the number of encoder pulses counted per window. The speed change ($-4\,\%$) is of the same order.

- **Missing factors**: one run per case; wheel inertia not measured; gear ratio and driver stop mode (coasting or active braking) not recorded; sampling step of about 0.1 s, close to τ_1; time constants read to two decimals, so the $+64\,\%$ could lie anywhere between about $+52$ and $+76\,\%$.

This test shaped what followed: study the flywheel shape first, then braking on a simpler set-up.

3 TIPE, part 1: flywheel shape

3.1 Problem statement

The kinetic energy of a solid rotating about a fixed axis is $E_c = \frac{1}{2} I\omega^2$. At a given speed, maximising E_c means maximising $I = \int (x^2 + y^2)\,\mathrm{d}m$. The flywheel is assumed homogeneous and balanced: its centre of mass lies on the axis, and the axis is a principal axis of inertia. To compare shapes rather than sizes, one quantity is fixed (perimeter, area or volume).

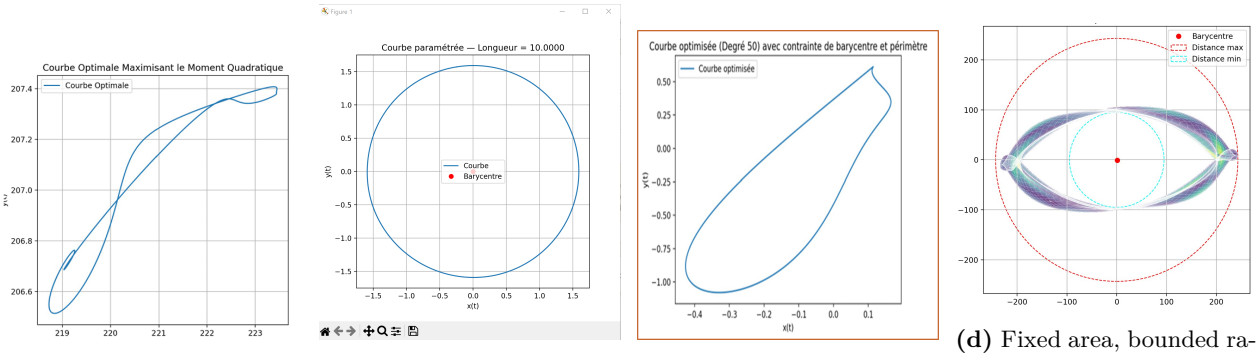
3.2 Plane curve at fixed perimeter

The curve is parametrised by truncated Fourier series,

$$x(t) = a_0 + \sum_{n=1}^{N} \big(a_n \cos 2\pi n t + b_n \sin 2\pi n t\big), \qquad y(t) = c_0 + \sum_{n=1}^{N} \big(c_n \cos 2\pi n t + d_n \sin 2\pi n t\big),$$

and $M = \int_0^1 (x^2 + y^2)\sqrt{x'^2 + y'^2}\,\mathrm{d}t$ is maximised subject to $L = L_0$.

- With no other constraint, the optimiser simply moves the curve away from the origin (figure 2a). The centroid must therefore be fixed at the origin.
- With the *trust-region constrained* method, which enforces the constraints strictly, the result is a circle (figure 2b). Check: for $L_0 = 10$, $R = L_0/2\pi \approx 1.59$ and $M = 2\pi R^3 \approx 25.3$, the value found by the optimiser.
- A polynomial parametrisation, $x(t) - x(0) = t(t-1)\sum a_i t^i$ (closed by construction), gives less regular curves. At high degree, spurious oscillations appear near the ends (Runge's phenomenon, figure 2c). Fourier series are therefore preferable.



(a) No centroid constraint **(b)** Fourier, trust-constr **(c)** Polynomial (degree 50) **(d)** Fixed area, bounded radius

Figure 2: Shape optimisation in the TIPE (original plot labels in French).

3.3 Plane surface at fixed area, and the 3D case

At fixed area the problem has no solution: an annulus of ever larger radius and ever smaller thickness always increases I, so the computation does not converge. The TIPE added a maximum and a minimum radius (polar parametrisation $r(t)$). The resulting shape oscillates between the two bounds without reaching a clean solution (figure 2d). An approach based on a parametrised 3D surface was described, without results. Both leads were left unfinished. Part 1 settles them exactly (section 6).

4 TIPE, part 2: braking

4.1 Model

For a vehicle in plane motion subject to drag, braking force and tyre friction (figure 3a),

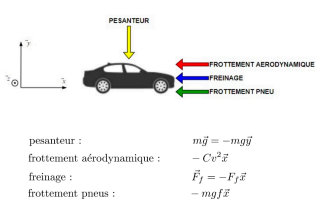
$$-\tfrac{1}{2}C_x\rho S\,v^2 - F_f - mgf = m\frac{\mathrm{d}v}{\mathrm{d}t} \;\Rightarrow\; v(t) = \gamma\tan\Big(\arctan\frac{v_0}{\gamma} - \frac{\alpha}{m\gamma}t\Big),$$

with $\alpha = F_f + mgf$ and $\gamma = \sqrt{\alpha/(\tfrac{1}{2}C_x\rho S)}$. This gives the stopping time $t_f = \frac{m\gamma}{\alpha}\arctan\frac{v_0}{\gamma}$ and, by integration, the distance $X(t_f)$.

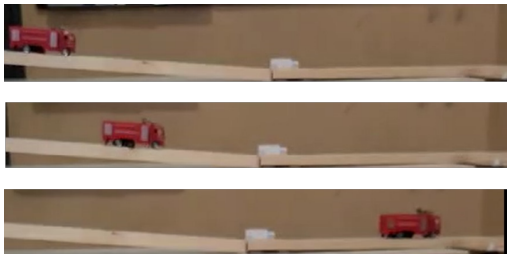
Compared with the French driving-school rule $d = (v_{\mathrm{km/h}}/10)^2$, the theoretical distance is much shorter at 50 km/h: 11.8 m according to the TIPE table (12.3 m with the deceleration gf alone, $f = 0.8$) against 25 m. The gap does not come from drag, which is negligible at these speeds. It comes from the assumed deceleration: $gf \approx 7.8$ m/s² here, against about 3.9 m/s² implied by the rule (computed for this report).

4.2 Experiment

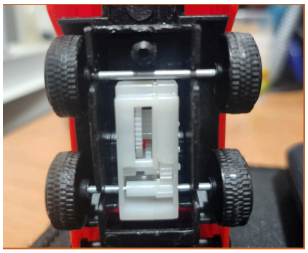
- **Vehicle**: a 34 g toy truck (frontal area 11.55 cm²), released at rest at the top of a ramp, then slowed by friction on a horizontal section.
- **Measurement**: the motion is filmed and tracked with Tracker; speeds are obtained by finite differences.
- **Friction**: f is measured on an inclined plane at the onset of sliding, $f = \tan\theta = \tan 5.89° \approx 0.10$.
- **Drag**: C_D is estimated from the Reynolds number ($Re \approx 10^3$, $C_D \approx 1.1$).
- **Flywheel**: mimicked by a torsion spring linked to the wheels through a high-ratio gear train, using the analogy between elastic and kinetic energy (figure 3c).



(a) Forces



(b) Filmed sequence



(c) Spring and gear train

Figure 3: Braking study in the TIPE.

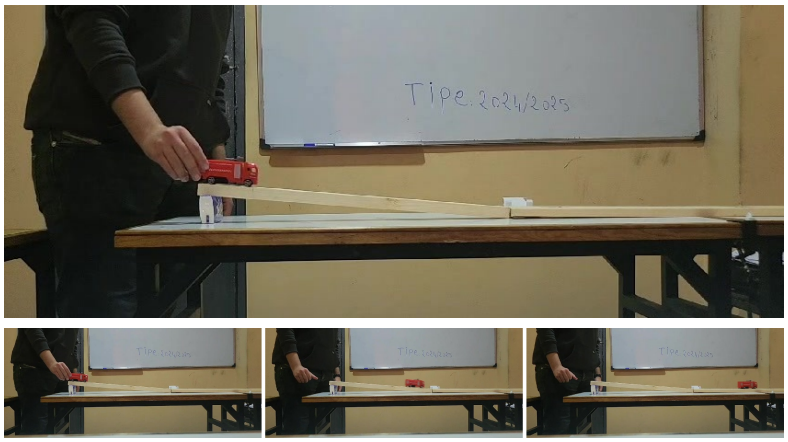


Figure 4: The experiment in progress (February 2025): releasing the truck at the top of the ramp, then descent and braking on the horizontal section.

4.3 Results presented in the TIPE

	Theoretical (m)	Measured (m)
Without flywheel	0.87	0.82
With flywheel (spring)	1.10	1.08

Table 3: Braking distances, final TIPE presentation.

The agreement between model and measurement was presented as a validation of the braking model.

5 Critical re-analysis of the TIPE data

The Tracker files were re-analysed (figure 5). The values below come from that re-analysis.

- **Both vehicles are still moving in the last frame** (0.38 and 0.62 m/s). The recorded distances are therefore not stopping distances.
- **Entry speeds differ**: 1.32 m/s without flywheel against 1.49 m/s with. The initial kinetic energy is 28 to 47 % higher in the run with the flywheel, depending on the speed estimator, which is enough to explain the difference in distance.
- **Decelerations are equal within uncertainty**: 0.922 ± 0.020 m/s² without flywheel and 0.896 ± 0.015 m/s² with, i.e. apparent friction coefficients of 0.094 and 0.091. No effect of the spring is detectable.
- **Only one run per configuration** was kept.

Conclusion. The deceleration model is consistent with the measurement ($f \approx 0.09$–0.10). The effect of the flywheel on braking, however, is not demonstrated: the two runs do not have the same entry speed, each was done only once, the recording stops before the vehicle does, and the energy taken up by the spring is not measured. The difference between the two decelerations (0.026 m/s²) is about one combined standard deviation.

Two physical caveats. The coefficient measured on the inclined plane at the onset of sliding is not directly a rolling resistance (wheel and axle friction). The spring stores elastic energy, not rotational kinetic energy: their torque–angle and torque–speed laws differ, so the spring test does not directly validate a flywheel recovery system.

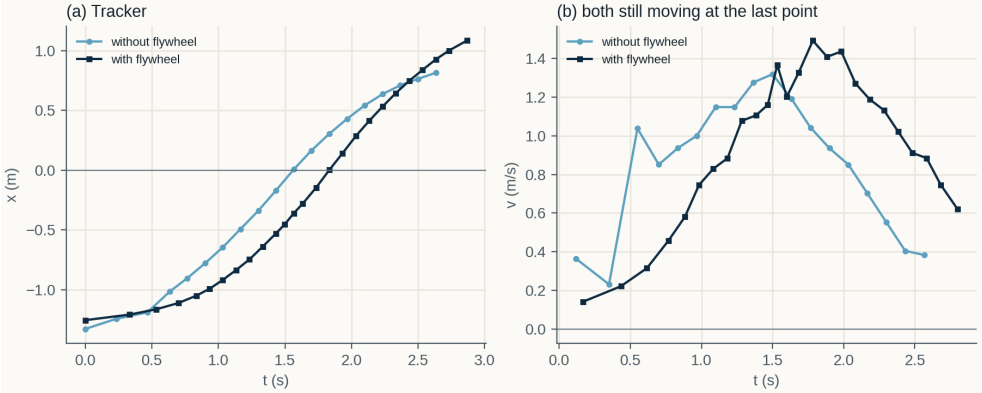


Figure 5: Re-analysis of the Tracker data.

6 Part 1: proving the shape result

Both shape results are proved (the circle had been found numerically in the TIPE; the fixed-area case had not converged):

- **Fixed perimeter and centroid**: $M \leq L^3/4\pi^2$ (Wirtinger's inequality), with equality for the circle.
- **Fixed area inside a disk of radius** R: the outer annulus maximises J, with $J_{\max} = AR^2 - A^2/2\pi$ (radial rearrangement).

The second result explains why the TIPE computation did not converge: without a maximum radius, there is no optimum. It also confirms the starting intuition of placing the mass as far as possible from the axis. Both theorems maximise I *at fixed speed*.

7 Part 2: the shape that stores the most before failure

7.1 A change of objective

A real flywheel is limited not by its speed but by the strength of its material. In a thin ring of radius R, the stress is $\rho\omega^2 R^2$ whatever its thickness, so the ring cannot be enlarged indefinitely. The right quantity to maximise is the specific energy at the allowable stress,

$$\frac{E}{m} = K\,\frac{\sigma_{\text{adm}}}{\rho}, \qquad \Lambda = \frac{\rho\,\omega_{\max}^2\,b^2}{\sigma_{\text{adm}}},$$

where K is a dimensionless shape factor and Λ characterises the speed limit. Two bounds apply: $K \leq \Lambda/2$ (all the mass at the rim) and $K \leq 1$ (twice the thin ring). These bounds and the existence of several speed regimes are known (Berger and Porat, 1988 [1]). Berger and Porat do not provide for a central bore [2]; the closest 1D/2D comparison is Kress's [3], and the most complete shape optimisation is that of Kale et al. [4]. What is added here is a single-parameter map, the exact solution of the speed-limited regime, and the effect of the bore and of the minimum web thickness.

Shape	Shape factor K
Thin ring (TIPE optimum)	0.5
Uniform-thickness disk, solid (no bore, not comparable to the others)	0.61
Uniform-thickness disk with bore ($a = 0.1\,b$)	0.31
Slope-limited profiles of Part 3, checked in 2D	0.57 to 0.67
Upper bound (constant-stress disk)	1

Table 4: Shape factors (isotropic material, $\nu = 0.3$).

7.2 Main results

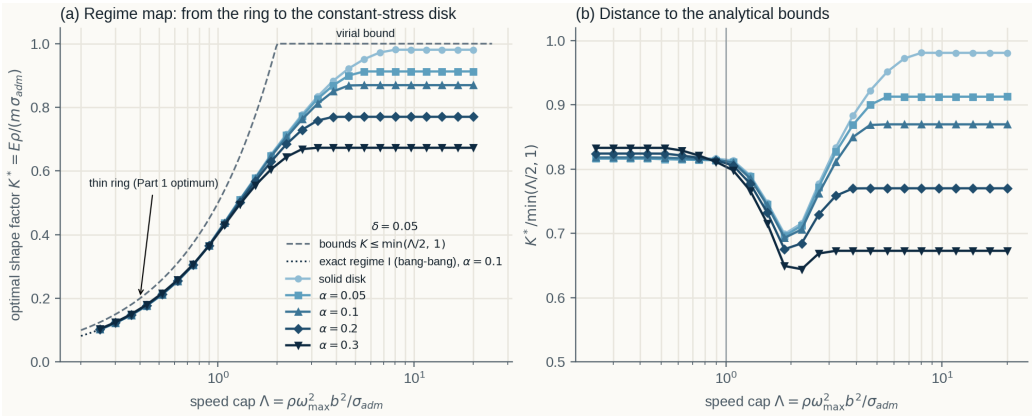
- **Three regimes** (figure 6a). At low Λ (speed-limited), the optimum is the ring of Part 1, with an exact "rim on web" solution. At high Λ (strength-limited), it tends to a constant-stress disk (Stodola type).
- **1D optima fail in 2D without a slope limit.** The optimal profile of the 1D (plane stress) model has near-vertical steps at the hub. A 2D axisymmetric computation reveals a stress concentration there that grows as the mesh is refined. Limiting the slope to $\mathrm{d}h/\mathrm{d}r \approx 0.25$–$0.5$ recovers $K = 0.70$–0.82 in 2D, depending on the slenderness $h_{\max}/b$ (0.2 or 0.05) (figure 6b). The Part 3 parts, with printable webs of 1 and 2 mm, give 0.67 and 0.57. The slope limit alone is not enough: an abrupt change of slope is still a re-entrant corner. The profiles are therefore smoothed by monotone interpolation.

  Mesh convergence. The 2D check of the three printed parts was repeated on five meshes, each twice as fine as the previous one in both directions (table 5). For all three parts, the maximum stress is at the edge of the bore, on the mid-plane: this is the usual maximum of a bored disk, not a geometric singularity. Between the two finest meshes, K changes by 0.13 % (uniform disk, 2 mm web) and 0.29 % (1 mm web). For the 1 mm web the change no longer shrinks between the last two refinements, so the retained value is $K = 0.67 \pm 0.01$. The ratio to the reference part, which is what the profile comparison uses, changes by less than 0.2 % over the five meshes.

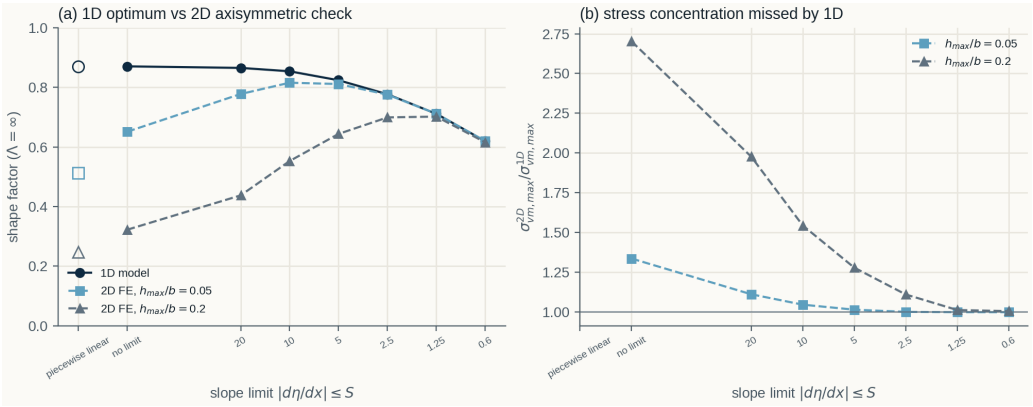
Mesh ($n_r \times n_z$)	75×3	150×6	300×12	600×24	1200×48	Ratio to uniform
Uniform, bored	0.311	0.308	0.306	0.306	0.305	1
1 mm web	0.688	0.681	0.678	0.676	0.674	2.21
2 mm web	0.582	0.576	0.573	0.572	0.571	1.87

Table 5: Shape factor K_{2D} of the printed parts versus mesh (slenderness 0.2, $\nu = 0.3$).

- **The bored uniform-thickness disk is dominated**: a profile filling 84 % of the envelope beats it in both energy per kilogram and energy per envelope volume, i.e. per volume of the cylinder of radius b and height $h_{\max}$ (+32 % and +11 %; $1.32 \times 0.84 \approx 1.11$). Per volume of material, the gain would equal the per-kilogram gain.
- **The price of robustness**: yield strength is drawn with a 5 % coefficient of variation, and the thickness receives a Gaussian error field (standard deviation 0.5 % of $h_{\max}$, correlation length $0.1\,b$). The safety factor SF is the one that brings the probability of yielding down to 1 in 1000 (20,000 Monte Carlo draws), and $K_{\mathrm{eff}} = K/\mathrm{SF}$. The nominal optimum with the thinnest web needs SF = 2.25 ($K_{\mathrm{eff}} = 0.40$). A profile re-optimised under uncertainty (sample-average approximation, checked on fresh draws) reaches $K_{\mathrm{eff}} = 0.684$.



(a) Regime map



(b) 2D check

Figure 6: Part 2: optimal shape factor as a function of Λ, and 1D/2D comparison.

7.3 Why a profiled disk and not a hollow cylinder

- **Balance**: the rotors are solids of revolution, hence statically and dynamically balanced by construction. These are the two conditions set in the TIPE.
- **Performance**: for an isotropic material (metal, PLA), the profiled disk beats the ring (K from 0.57 to 0.67 against 0.5). Composite flywheels are rings because their fibres are strong mainly in the hoop direction, which is outside this framework.
- **Manufacturing**: a profile of revolution is made on a lathe. The slope limit comes from the 2D check; the 1 mm minimum thickness is imposed by 3D printing (Part 3).

7.4 The criterion matters

The shape factor K measures energy *per kilogram*. Other criteria give a different ranking. For the three Part 3 parts (same material, same b):

Rotor	Mass (g)	K_{2D}	Total energy before failure ($\propto Km$)	Energy at equal speed ($\propto I = jmb^2$)
Uniform reference (4 mm)	38.6	0.306	100 %	100 %
1 mm web	14.5	0.67	84 %	31 %
2 mm web	27.7	0.57	134 %	66 %

Table 6: Ranking by criterion (2D model, no other speed limit; computed for this report).

The 1 mm web is the best per kilogram but the worst in total energy before failure. At equal speed, both profiles store less than the heavier reference. The right criterion depends on the application: energy per kilogram (on-board use), total energy within a given envelope, or energy at an imposed operating speed. The trade-off between the first two is the Pareto front of Part 2.

8 Part 3: design for 3D printing

8.1 Parts

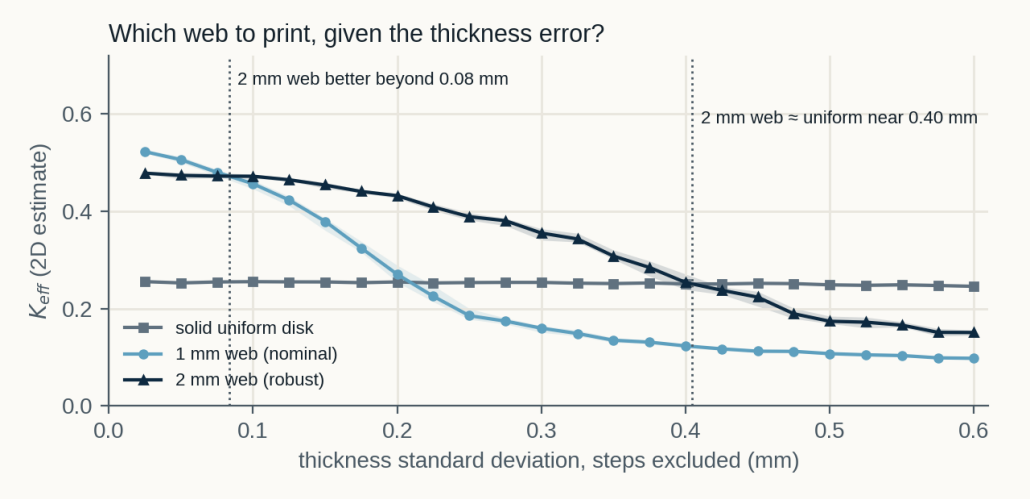
Three rotors of radius $b = 50$ mm are designed to be 3D printed flat as half-profiles: flat bottom face, profile $h(r)$ on top (figure 9). For a homogeneous material, mass and polar inertia depend only on $h(r)$, so the half-profile has the same inertia ratio $j = I/(mb^2)$ as the symmetric rotor, without print supports. The three parts are:

- a uniform-thickness reference (4 mm), whose $j = 0.505$ is known exactly;
- a nominal optimum with a 1 mm web ($K_{2D} = 0.67$);
- a robust optimum with a 2 mm web ($K_{2D} = 0.57$), both slope-limited ($\mathrm{d}h/\mathrm{d}r \leq 0.5$).

Model limits. K_{2D} is computed for the symmetric rotor, not for the printed half-profile (one flat face, one profiled face), whose stress field may differ. FDM-printed PLA is anisotropic (properties depend on layer and bead orientation), whereas the model assumes an isotropic material. The printed parts are therefore meant to check geometry and mass distribution, not strength.

8.2 Modelling results

- **Optima without a slope limit fail in 2D before any print**: $K = 0.28$, below the uniform disk (0.306). This is why the parts are slope-limited.
- **Layer steps are a sensitive point**: if the 0.2 mm layers are modelled as perfectly sharp steps, the 2D stress peak does not converge and K falls back to the uniform disk's level. A real layer is not a sharp step (rounded beads, defects), so the only conclusion is that the model is very sensitive to the surface state. A rotor meant to spin must therefore have its surface finished.
- **The choice of web depends on the printed thickness error** (figure 7). The 1 mm web is only better for a scatter below about 0.08 mm, and the uncertainty on the 2D factor does not allow it to be preferred with confidence. The 2 mm web wins beyond about 0.13 mm: it keeps 1.7 times the uniform disk's effective factor at 0.2 mm and 1.4 times at 0.3 mm, and meets it near 0.40 mm.
- **Trifilar pendulum designed** to measure the inertia ratio: calibrated on the uniform part (which cancels g, wire length and radius), it separates the two profiles ($j = 0.420$ vs 0.462) by about 9 standard uncertainties with a stopwatch and 34 with 60 fps video (budget: timing, masses and radii).



(a) Choice of web vs thickness error

Figure 7: Part 3: choice of web versus printed thickness error.

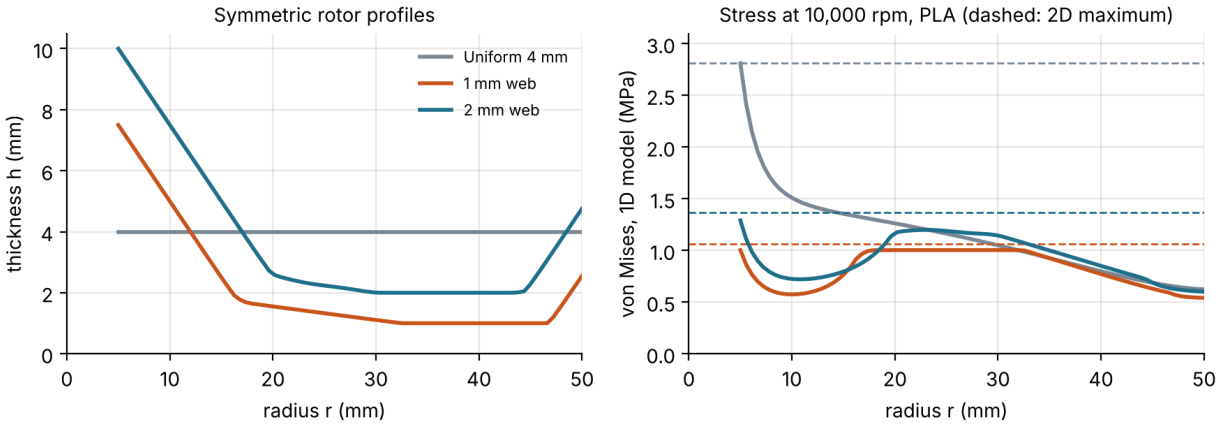


Figure 8: Profiles of the three symmetric rotors and von Mises stress computed along the radius (1D model); dashed: 2D maximum.

9 CAD and finite-element simulation

9.1 CAD

- **CATIA 3DEXPERIENCE**: the Part 2 profiles are exported as (r, z) points with a protocol (spline, revolution, mass check).
- **Fusion 360**: a script builds the three printable parts from the optimised profiles. Volume difference with the Python model: 0 % for the reference, −0.05 % and −0.03 % for the profiles, due to spline smoothing.

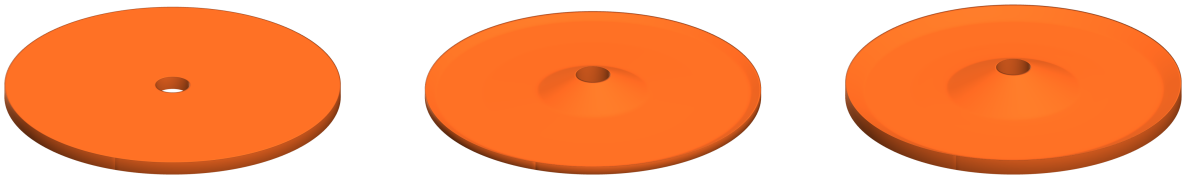


Figure 9: The three printable parts at the same scale: uniform reference, 1 mm web, 2 mm web (rendered from the profiles used in Fusion 360).

9.2 Abaqus input files checked with CalculiX

Abaqus input files (`.inp`) are generated for four steel rotors with $b = 150$ mm: CAX8R axisymmetric elements, centrifugal loading, one version under 1000 nodes for the student licence and a fine version with 11,425 nodes. They were run with CalculiX 2.21, a free solver that reads the Abaqus format. In each case the speed is chosen so that the maximum stress of the 1D model is 574.7 MPa.

Rotor	1D	In-house FE	CalculiX (fine)
Uniform disk with bore	574.7	572.7	570.2
Slope-limited profile, $\Lambda = \infty$	574.7	635.3	630.4
Slope-limited profile, $\Lambda = 2$	574.7	620.0	616.8
1D optimum without slope limit	574.7	1397.7	931.5 then 1405.1*

Table 7: Maximum von Mises stress (MPa). *Student then fine mesh: the peak grows with the mesh, the sign of a singularity at the hub.

CalculiX and the in-house code agree to 0.4–0.8 %.

9.3 Stresses in the printed parts

At 10,000 rpm in PLA ($\rho = 1240$ kg/m^3, $\nu = 0.3$), the maximum von Mises stress is 2.81 MPa for the reference, 1.06 MPa for the 1 mm web and 1.37 MPa for the 2 mm web (2D model), far below the strength of PLA. Figure 10 shows the distribution computed with the 1D model.

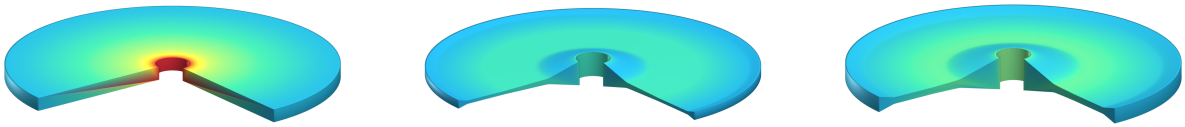


Figure 10: Von Mises stress at 10,000 rpm in PLA computed with the 1D model, same scale for the three parts (0 to 2.81 MPa, blue to red).

10 Summary and limitations

Established.

- The circle (fixed perimeter) and the annulus (fixed area and radius) maximise inertia: proved.
- The deceleration model is consistent with the measurement.
- Once failure is taken into account, a slope-limited profiled disk beats both the ring and the bored uniform disk: computed in 1D and 2D, confirmed with CalculiX on the Part 2 rotors.
- The profiles adapted to printing keep this gain: $K = 0.67$ and 0.57 against 0.31 for the bored uniform disk, stable over five meshes.

What the data cannot conclude. The effect of the flywheel on braking (section 5).

Limits of the framework. Isotropic, homogeneous material; no composites, fatigue or rotor dynamics; typical rather than measured material data; losses (bearings, air) not modelled.

A Notation and relations

Symbol	Meaning
b, a	outer radius and bore radius ($a = 0.1\,b$ throughout)
$h(r)$	thickness of the symmetric rotor at radius r
ρ, ν, σ_{adm}	density, Poisson's ratio, allowable stress
$E_c = \frac{1}{2} I \omega^2$	rotational kinetic energy
$K = \dfrac{E\,\rho}{m\,\sigma_{\mathrm{adm}}}$	shape factor: specific energy relative to $\sigma_{\mathrm{adm}}/\rho$
$\Lambda = \rho\,\omega_{\max}^2 b^2/\sigma_{\mathrm{adm}}$	dimensionless speed limit
$j = I/(mb^2)$	inertia ratio
$\sigma_\theta = \rho\omega^2 R^2$	stress in a thin ring of radius R (hence $K = 0.5$)
$M \le L^3/4\pi^2$	second moment of a centred curve of perimeter L (Wirtinger)
$J_{\max} = AR^2 - A^2/2\pi$	second moment of a region of area A inside a disk of radius R

B Manufacturing

Printing (same settings for every part): flat face on the bed, no supports, no brim or raft; fixed 0.2 mm layer height, first layer included; 100 % infill, concentric pattern, 3 perimeters; same PLA spool. The rotor bore is 10 mm in diameter ($a = 5$ mm).

Part	Volume (cm³)	PLA mass (g)	Size (mm)
Uniform reference	31.1	38.6	$100 \times 100 \times 4$
1 mm web	11.7	14.5	$100 \times 100 \times 7.5$
2 mm web	22.4	27.7	$100 \times 100 \times 10$

Table 8: Rotors designed for printing (masses computed with 1.24 g/cm³).

C Tools

Tool	Use
Python (NumPy, SciPy)	shape optimisation (TIPE), 1D rotating-disk solver, constrained optimisation, Monte Carlo
scikit-fem	2D axisymmetric finite elements (in-house code)
CalculiX 2.21	check of the Abaqus-format files
Fusion 360	CAD of the printable parts
Tracker	video tracking of the braking experiment
Arduino	motor bench of the preliminary test

Code and result files are available on request.

References

[1] M. Berger, I. Porat, J. Appl. Mech. 55(1):164–170, 1988. doi:10.1115/1.3173623

[2] R. Okou et al., IEEE Trans. Energy Convers. 26(4):1198–1209, 2011. doi:10.1109/TEC.2011.2160545

[3] G.R. Kress, *Shape optimization of a flywheel*, Struct. Multidisc. Optim. 19:74–81, 2000. doi:10.1007/s001580050087

[4] V. Kale, M. Thomas, M. Secanell, Struct. Multidisc. Optim. 64:1481–1499, 2021. doi:10.1007/s00158-021-02935-x

[5] M. Hedlund et al., Energies 8:10636–10663, 2015. doi:10.3390/en81010636